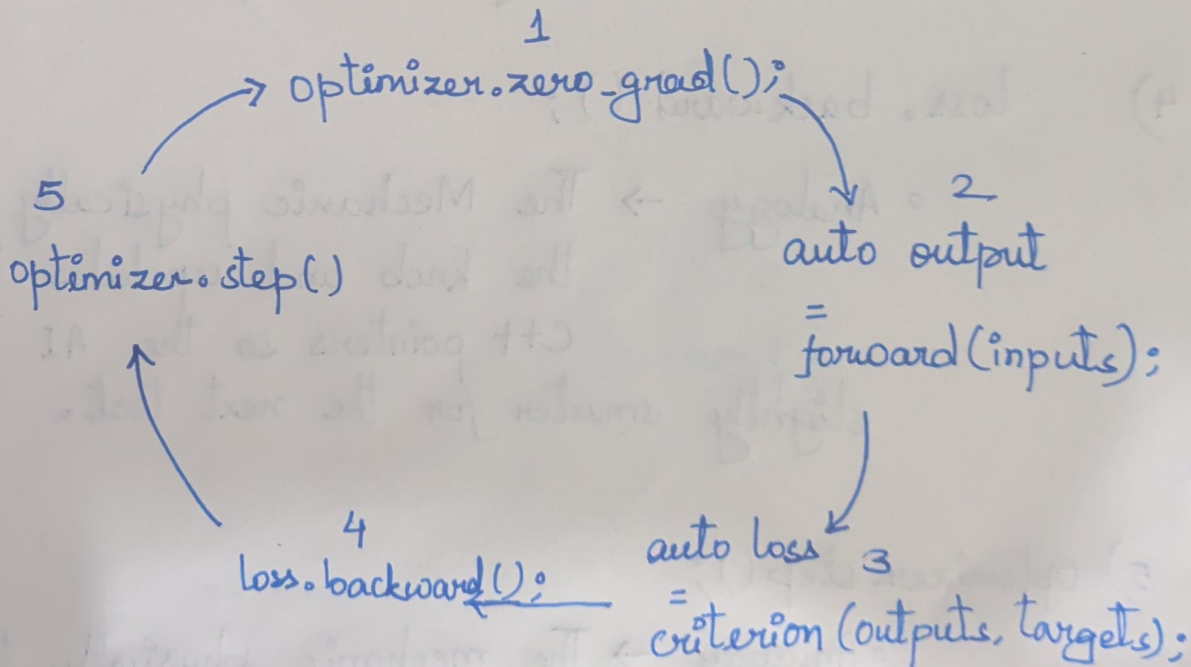


LibTorch Day 9

Two Core Engines

- Loss Function (`torch::nn::CrossEntropyLoss`): The Grader. It calculates exactly how wrong the AI's prediction was compared to the true answer. Lower number = better.
- Optimizer (`torch::optim::Adam`): The Mechanic. It looks at the errors and physically adjusts the mathematical weights inside the network to make the next guess better.

The 5-step Learning Cycle



1) `optimizer.zero_grad()`;

- Analogy $\rightarrow$ like wiping the chalkboard clean before solving a new math problem. If you forget this, the old math mixes with the new math and the network crashes.

2) `auto output = forward(inputs)`;

- Analogy $\rightarrow$ The AI takes a blind test and guesses the answers.

3) `auto loss = criterion(output, targets)`;

- Analogy $\rightarrow$ The teacher grades the test and returns a score of how badly the AI failed.

4) `loss.backward()`;

- Analogy $\rightarrow$ The Mechanic physically turns the knob and updates the C++ pointers so the AI is slightly smarter for the next test.

5) `optimizer.step()`;

- Analogy $\rightarrow$ The mechanic physically turns the knobs and updates the C++ pointers so the AI is slightly smarter for next test.